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Project Title

Structural mapping of the *in vivo Pseudomonas aeruginosa* biofilm matrix exopolysaccharide, Psl, does it change with virulence?

Summary

Quantitative 3D mapping of Psl exopolysaccharide in 72 murine tissue samples using IHC, CLSM and automated ZEISS Intellesis segmentation to correlate biofilm structure with genotypic virulence.

My Role: Lead Researcher - designed wet-lab pipeline, imaging workflow and automated image-analysis.

Methods

Sample prep	Tissue fixation, embedding, H&E, IHC staining, PA01 antibody, HHL-TRITC, POPO1.
Imaging	Confocal Microscopy (CLSM), ZEISS Axioscan Slide scanner, fluorescent, brightfield & polarised
Image Analysis	ZEISS Intellesis machine-learning segmentation pipeline; and metrics quantification
Stats and Plots	Data engineering and statistics in R and GraphPad Prism (Two-way RM ANOVA, post-hoc tests, correlation tests).

Key Results

• Processed and analysed **72 tissue samples** with a consistent imaging pipeline. Automated segmentation reduced manual processing time by **~70%** and produced reproducible volume metrics (CV < 8%).
• Statistically significant correlation between Psl volume and virulence genotype ($p < 0.01$). Biofilm/ aggregate is not solely dependent on Psl expression levels.

Impact / Takeaway

Provided a reproducible, high-throughput imaging + analysis pipeline enabling data-driven comparisons across genotypes - directly translatable to assay validation or phenotypic screens.

Images and Plots

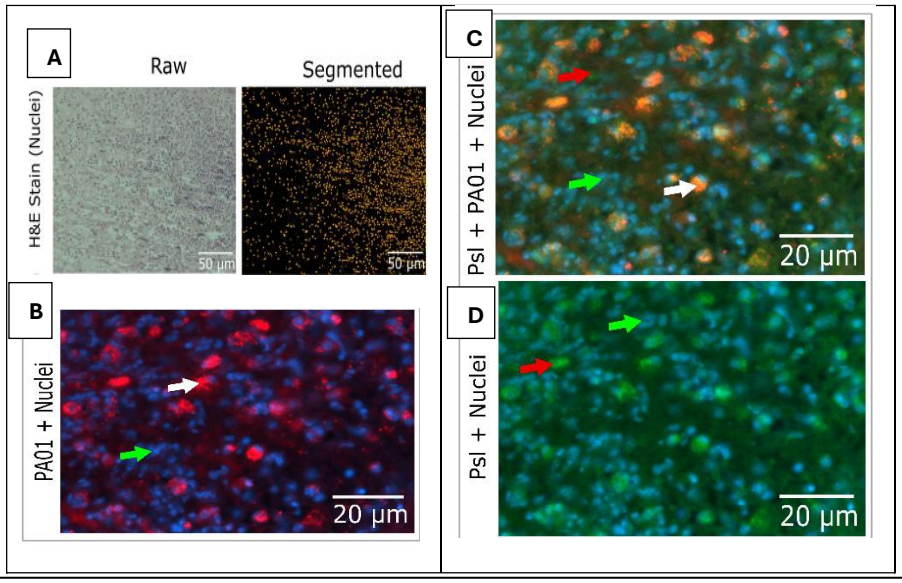


Figure X. (A) Representative ROIs stained with H&E and acquired using ZEISS Axioscan 7 slide scanner 20× objective. Raw images were segmented and analysed in ZEISS Intellesis.

(B-D) Representative image (CE group) showing bacterial aggregates (white arrows), Psl clusters (red arrows) and host nuclei (POPO-1, green arrows). Acquired on ZEISS Axioscan 7, 20×; scale bar = 20 µm.

Available for discussion. Full Psl quantitative and reproducible analysis: [GitHub link or Portfolio PDF]

Abstract

Pseudomonas aeruginosa forms complex biofilms that contribute to multidrug resistance, immune evasion, and infection persistence in clinical practice. The biofilm extracellular matrix is primarily composed of exopolysaccharides; alginate, Pel and Psl, that contribute to its architectural integrity. Despite the development of biofilm-targeted therapeutics, highly virulent strains that overexpress Psl remain prevalent, contributing to nosocomial infections with poor strains that overexpress Psl remain prevalent, contributing to nosocomial infections with poor strains that overexpress Psl remain prevalent, contributing to nosocomial infections with poor treatment outcomes. While past studies have demonstrated the role of Psl in biofilmdevelopment and maintenance, the relationship between *in vivo* Psl spatial distribution and virulence outcomes remains unclear. In this study, we aim to produce *in vivo* structural maps of Psl, and link its spatial distribution and localisation to genotypic virulence, and immunomodulation. We investigate Psl spatial distribution in 12 BALB/c mice using three genetic variants of *Pseudomonas aeruginosa* PA01 strain; wild-type (WT), constituitive expression (CE), and deletion. A machine learning-based tool;genetic variants of *Pseudomonas aeruginosa* PA01 strain. A machine learning-based tool; ZEISS Intellesis, is used for high-throughput analysis of 3D images acquired using ZEISS Axioscan 7 slide scanner. We show that (i) Psl levels are elevated in highly virulent strains; (ii) biofilm/aggregate formation is multifactorial rather than entirely dependent on Psl expression levels (iii) bacterial aggregates impede host immune cell penetration and phagocytic-killing. In summary, these findings suggest that Psl is directly related to virulence, and it contributes to biofilm/aggregate formation, thereby making it a target for the development of efficacy therapeutic interventions against biofilm infections.

Futher Work

- Future work should therefore use thoroughly trained Intellesis models, and ZEISS Arivis Cloud which has more advanced deep learning features.
- Comparing Psl and Pel in the same murine models, and extending the study to humanskin/wound infections, will further elucidate clinically relevant biofilm-host interactions.
- Because Psl reinforces biofilms and modulates immune responses and antibiotic tolerance, it presents as a crucial anti-biofilm therapeutic target. Future studies should focus on novel anti-Psl strategies that disrupt biofilms hence tackle antimicrobial tolerance and improve infection outcomes.